

UNIX System Services





OMVS OPENS THE USS COMMAND SHELL

IBM

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```
IBMUUSER : /u/ibmuser : > uname -a
```

```
OS/390 VS01 29.00 05 8562
```

```
IBMUUSER : /u/ibmuser : > ssh -V
```

```
OpenSSH_8.4p1, LibreSSL 3.0.2
```

```
IBMUUSER : /u/ibmuser : > python --version
```

```
IBM Open Enterprise SDK for Python 3.13.8
```

```
IBMUUSER : /u/ibmuser : > javac -version
```

```
javac 11.0.22
```

```
IBMUUSER : /u/ibmuser : >
```

The **omvs** command at the TSO READY prompt or on ISPF Option 6 opens the USS command line shell.

The screenshot shows the output from `uname` and the versions of `ssh`, `python`, and `javac` that are installed.

You can also access your command line shell by connecting with `ssh`.

THE OPEN GROUP: Z/OS CERTIFIED UNIX 95



Although UNIX System Services is not a standalone UNIX operating system, it meets the certification standards of The Open Group. z/OS 3.1 and later is certified UNIX 95.

USS POSIX COMPLIANCE

z/OS UNIX System Services is *generally* POSIX-compliant, but because it makes use of MVS components of z/OS and must interact and run in the MVS environment, there are some divergences from the standard. z/OS rather than USS is considered "the system."

USS maps POSIX calls to z/OS services. Timing-related syscalls, boundary conditions and edge cases, and errno behavior can diverge from the standard.

USS may not implement all optional parts of POSIX interfaces.

USS maps permissions, link, ownership, and filesystem semantics to z/OS facilities.

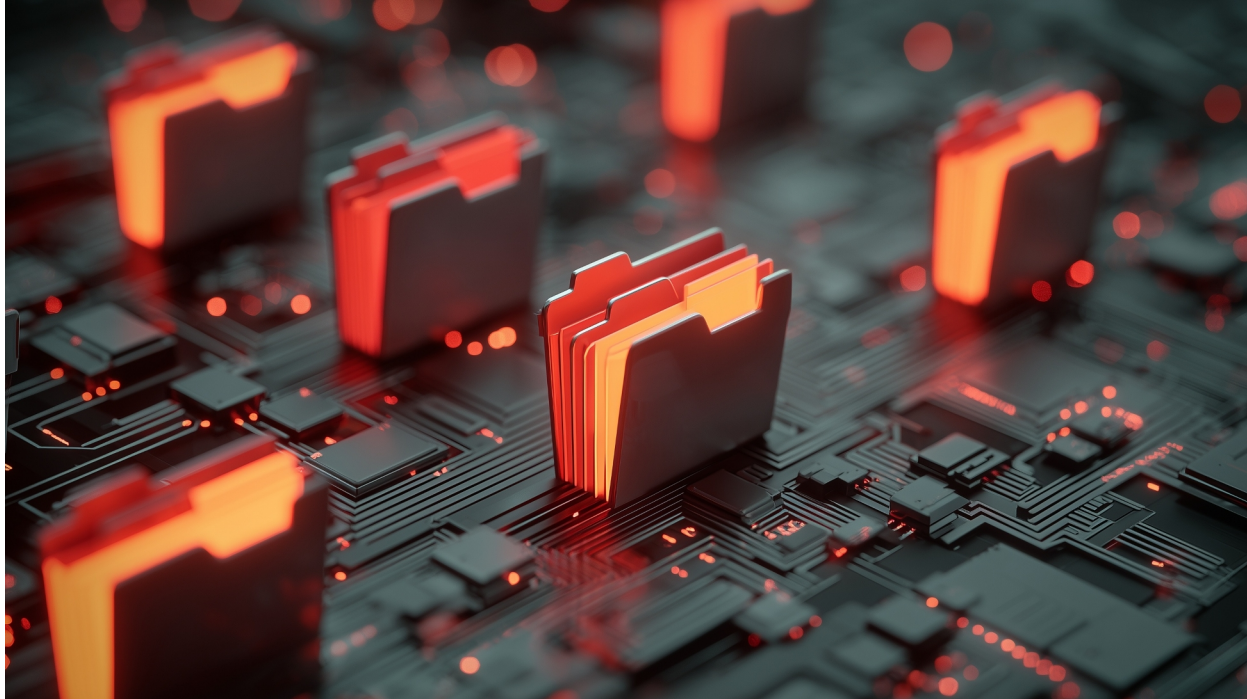
USS processes run in z/OS address spaces. Signal delivery and process / thread / thread group behavior can vary from POSIX.

USS inherits z/OS resource constraints and may have different limits for things like queue sizes, path length, and file descriptors.

pthread APIs map to z/OS task/subtask (thread) implementation and behavior. Thread cancellation, scheduling, and synchronization corner cases behave differently from POSIX.



USS FILESYSTEMS



Unretouched photo of an actual filesystem in the wild

USS filesystems on z/OS 3.2:

- zFS – the primary USS filesystem
- HFS – formerly the primary filesystem
- NFS – mount/export of NFS
- TFS – for transparent integrated filesystem support

zFS can run in a colony address space separate from the OMVS address space.

zFS can place stacks above the bar (addresses larger than 2GB).

Each filesystem is stored in a VSAM Linear Data Set (LDS). In USS terminology these data sets are called *aggregates*, because one of them can hold more than one USS filesystem.

UNIX ON IBM MAINFRAMES

1975-1979: A team at Princeton University ported UNIX to the IBM VM/370 operating system. One of them, Tom Lyon, joined Amdahl (a maker of IBM-compatible mainframes) in 1978 and by 1979 there was a working prototype of UNIX 6 running on an Amdahl 470. They continued development and had UNIX 7 running in 1979. Amdahl continued development until 1993, calling the product the Universal Timesharing System (UTS).

1979: IBM worked with Bell Labs to develop a version of UNIX for System/370, but it was not a general-purpose OS and was used internally at Bell as a build server for switches.

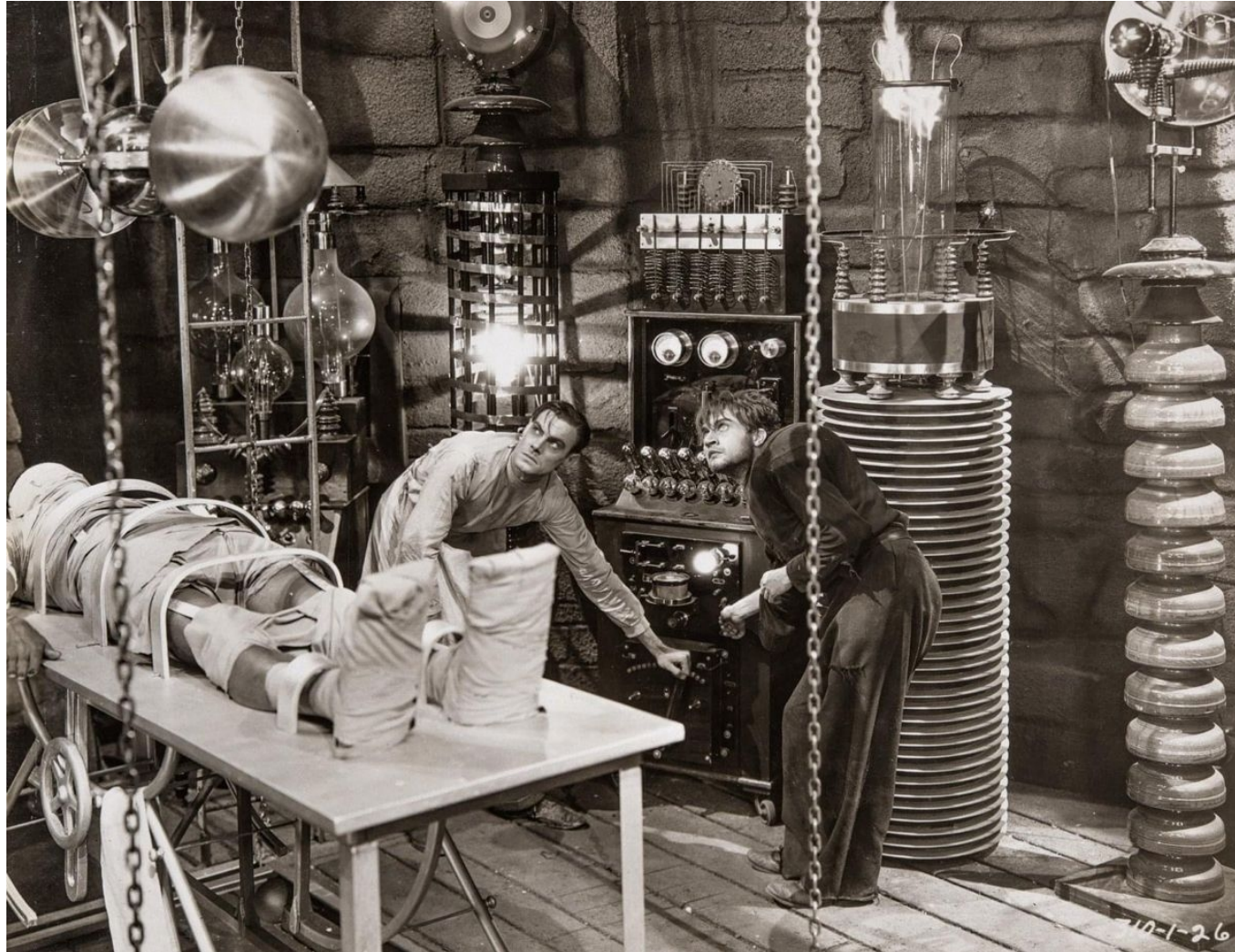
1985: IBM announced VM/IX, which they picked up from Interactive Systems Corporation (ISC). It was a UNIX III implementation that ran as a guest OS under IBM's VM. VM/IX did not become generally available.

1985: IBM and ISC released IX/370, a fully supported port of System V UNIX. The intent was to compete against Amdahl's UTS.

1991: Sun Microsystems released OpenSolaris and subsequently open-sourced most of the source code (2005). Sine Nomine began development of Solaris for the IBM mainframe in 2006, and delivered a series of demonstrations for IBM between then and 2009. After Oracle's acquisition of Sun Microsystems, they reverted the open source license to closed source, and the project was terminated in March of 2010.

1993: IBM introduced MVS Open Edition, a (mostly) POSIX-compliant optional component for the MVS/ESA SP 4.3 operating system (a direct ancestor of z/OS). The product evolved alongside the operating system and is now branded as UNIX System Services (USS).

LAB Z/OS 35: EXPLORE USS COMMANDS



WHAT YOU KNOW AND WHAT YOU CAN DO

- You know what USS is and where it came from
- You can access and use your USS command line shell

WHAT'S NEXT

- Software Installation and Management